# AI-Powered Enterprise E-Commerce Platform

# AI Model Architecture Specification

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# Document Information

| Item | Description |
|---|---|
| Project Name | AI-Powered Enterprise E-Commerce Platform |
| Document Type | AI Model Architecture |
| Version | 1.0 |
| Prepared By | Project Team |
| Status | Draft |

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# Revision History

| Version | Date | Description |
|---|---|---|
| 1.0 | DD/MM/YYYY | Initial Version |

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# Table of Contents

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# 1. Introduction

Artificial Intelligence is a major component of this platform.

The AI subsystem provides intelligent capabilities including:

- Personalized recommendations
- Fraud detection
- Sales forecasting
- Customer segmentation
- Conversational assistance

The architecture is designed to be modular, scalable, and deployable as independent microservices.

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# 2. Purpose

This document aims to:

- Define AI components.
- Specify model architectures.
- Define training pipelines.
- Establish deployment strategies.
- Support future model improvements.

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# 3. AI Objectives

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## AI-01

Improve customer experience.

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## AI-02

Increase sales through recommendations.

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## AI-03

Detect fraudulent activities.

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## AI-04

Provide business intelligence.

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## AI-05

Enable intelligent customer interactions.

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# 4. High-Level AI Architecture

```
Users
    ↓
Backend Services
    ↓
AI Gateway
    ↓
------------------------------------------------
| Recommendation Service                       |
| Fraud Detection Service                      |
| Forecast Service                             |
| Segmentation Service                         |
| AI Assistant                                 |
------------------------------------------------
```

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# 5. AI Services Overview

| Service | Purpose |
| --- | --- |
| Recommendation Service | Product recommendations |
| Fraud Service | Fraud risk detection |
| Forecast Service | Sales forecasting |
| Segmentation Service | Customer grouping |
| AI Assistant | Conversational support |

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# 6. AI Data Architecture

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# Data Sources

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## User Data

Examples:

- User profile
- Purchase history
- Browsing behavior

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## Product Data

Examples:

- Categories
- Ratings
- Prices

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## Transaction Data

Examples:

- Orders
- Payment history
- Fraud labels

---

## Historical Business Data

Examples:

- Sales history
- Seasonal trends

---

# Data Flow

```
Raw Data
     ↓
Cleaning
     ↓
Feature Engineering
     ↓
Training Dataset
```

```
     ↓
Models
```

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# 7. Recommendation System Architecture

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# Business Objective

Recommend relevant products to users.

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# Candidate Algorithms

### Phase 1

Popularity-Based Recommendations

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### Phase 2

Content-Based Filtering

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### Phase 3

Collaborative Filtering

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### Phase 4 (Future)

Hybrid Recommendation System

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# Recommended Initial Approach

Hybrid:

```
Content-Based
+
Collaborative Filtering
```

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# Inputs

- User history
- Product metadata
- Ratings
- Purchase history

---

# Outputs

Recommended product list.

---

# Pipeline

```
Customer Activity
       ↓
Feature Generation
       ↓
Recommendation Model
       ↓
Ranked Products
```

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# Evaluation Metrics

- Precision@K
- Recall@K
- MAP
- NDCG

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# 8. Fraud Detection Architecture

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# Business Objective

Identify suspicious transactions.

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# Inputs

- Payment amount
- Order frequency

- Device information
- User behavior

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# Candidate Algorithms

- Logistic Regression
- Random Forest
- XGBoost

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# Recommended Model

```
XGBoost
```

Reasons:

- High performance
- Handles imbalanced data well
- Good explainability

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# Output

Fraud Probability Score.

---

# Pipeline

```
Transaction Data
        ↓
Feature Engineering
        ↓
Fraud Model
        ↓
Risk Score
```

---

# Evaluation Metrics

- Accuracy

- Precision
- Recall
- F1 Score
- ROC-AUC

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# 9. Sales Forecasting Architecture

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# Business Objective

Predict future sales.

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# Inputs

- Historical sales
- Promotions
- Seasonality
- Holidays

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# Candidate Models

- Linear Regression
- ARIMA
- Prophet
- LSTM

---

# Recommended Initial Model

```
Facebook Prophet
```

Reasons:

- Easy implementation
- Handles seasonality
- Strong baseline performance

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# Future Enhancement

LSTM Models.

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# Outputs

Future sales predictions.

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# Evaluation Metrics

- MAE
- RMSE
- MAPE

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# 10. Customer Segmentation Architecture

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# Business Objective

Group similar customers.

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# Inputs

- Purchase frequency
- Spending behavior
- Product preferences

---

# Feature Framework

RFM Analysis:

- Recency
- Frequency
- Monetary Value

---

# Candidate Models

- K-Means
- DBSCAN

- Hierarchical Clustering

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# Recommended Model

```
K-Means Clustering
```

---

# Outputs

Customer segments.

Examples:

- High Value Customers
- Loyal Customers
- At Risk Customers

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# Evaluation Metrics

- Silhouette Score
- Davies-Bouldin Index

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# 11. AI Assistant Architecture

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# Business Objective

Provide conversational support.

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# Features

- Product questions
- Order tracking
- Recommendation assistance

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# Initial Implementation

Rule-Based Assistant.

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# Future Enhancement

RAG-Based Assistant.

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# Pipeline

```
User Query
      ↓
Intent Detection
      ↓
Response Generation
```

---

# 12. Feature Engineering Pipeline

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# Recommendation Features

- Purchase count
- User-product interactions
- Product categories

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# Fraud Features

- Transaction frequency
- Device usage
- Spending patterns

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# Forecast Features

- Time features
- Seasonal indicators
- Promotional indicators

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# Segmentation Features

- RFM metrics
- Behavioral statistics

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# Pipeline

```
Raw Data
     ↓
Cleaning
     ↓
Encoding
     ↓
Scaling
     ↓
Feature Store
```

---

# 13. Model Training Pipeline

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# Training Workflow

```
Data Collection
        ↓
Data Cleaning
        ↓
Feature Engineering
        ↓
Training
        ↓
Evaluation
        ↓
Model Registration
```

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# Training Frequency

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## Recommendation

Weekly.

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### Fraud

Monthly.

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### Forecasting

Monthly.

---

### Segmentation

Monthly.

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# 14. Model Storage

Store:

- Model files
- Metadata
- Evaluation reports

Recommended structure:

```
models/
│
├── recommendation/
├── fraud/
├── forecasting/
├── segmentation/
```

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# 15. Model Deployment Architecture

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# Deployment Approach

Independent AI microservices.

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# Technologies

- FastAPI
- Docker

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# Service Communication

```
Backend
     ↓
REST APIs
     ↓
AI Services
```

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# Deployment Diagram

```
Spring Boot
      ↓
Recommendation Service
Fraud Service
Forecast Service
Segmentation Service
```

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# 16. Model Monitoring

Monitor:

- Latency
- Error rates
- Accuracy degradation
- Data drift
- Resource usage

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# Metrics Examples

| Metric | Purpose |
| --- | --- |
| Prediction Latency | Performance |
| Accuracy | Model Quality |
| Drift Score | Dataset Changes |

| Metric | Purpose |
| --- | --- |
| Error Rate | Stability |

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# 17. Retraining Strategy

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# Triggers

- Scheduled retraining
- Accuracy degradation
- Data drift detection

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# Workflow

```
New Data
     ↓
Validation
     ↓
Training
     ↓
Evaluation
     ↓
Deployment
```

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# 18. AI Risks and Mitigation

| Risk | Mitigation |
| --- | --- |
| Poor Data Quality | Data Validation |
| Overfitting | Cross Validation |
| Model Drift | Monitoring |
| Bias | Fairness Evaluation |
| High Latency | Caching |

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# 19. AI Security Considerations

Protect against:

- Model extraction
- Data poisoning
- Adversarial attacks
- Unauthorized access

Security mechanisms:

- API authentication
- Dataset validation
- Monitoring
- Rate limiting

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# 20. AI Performance Targets

| Service | Target |
|---|---|
| Recommendation Latency | <5 sec |
| Fraud Detection Latency | <2 sec |
| Forecast Generation | <10 sec |
| Segmentation | Batch Processing |

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# 21. Future Enhancements

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## Recommendation

- Deep Learning Recommenders
- Reinforcement Learning

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## Fraud Detection

- Graph Neural Networks
- Real-Time Fraud Detection

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## Forecasting

- LSTM
- Transformer Models

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## AI Assistant

- RAG Architecture
- LLM Integration

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## MLOps

- MLflow
- Feature Store
- Automated Retraining Pipelines

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# 22. Recommended AI Directory Structure

```
ai-services/
│
├── recommendation/
├── fraud_detection/
├── forecasting/
├── segmentation/
├── assistant/
│
├── datasets/
├── notebooks/
├── models/
├── pipelines/
├── evaluations/
```

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# 23. Recommended Architecture Diagram

```
Users
   ↓
Backend
   ↓
AI Gateway
   ↓
---------------------------------
| Recommendation Service         |
| Fraud Detection Service        |
| Forecast Service               |
| Segmentation Service           |
| AI Assistant                   |
---------------------------------
```

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# 24. Conclusion

The proposed AI architecture provides:

- modular AI services,
- scalable deployment,
- enterprise-level model management,
- future MLOps readiness,
- and strong business intelligence capabilities.

The architecture enables gradual evolution from classical machine learning solutions to advanced AI systems while maintaining maintainability and scalability.

# References

[1] Géron, A. Hands-On Machine Learning.

[2] Ricci, F. Recommender Systems Handbook.

[3] Chollet, F. Deep Learning with Python.

[4] Prophet Documentation.

[5] XGBoost Documentation.

[6] Scikit-Learn Documentation.

[7] MLflow Documentation.

[8] Google MLOps Whitepaper.